

Adversarial Robustness of Traffic Light Recognition in Autonomous Vehicles

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I. INTRODUCTION

Autonomous vehicles (AVs) mainly depend on vision-based perception systems to detect and interpret traffic lights. Traffic-light recognition is especially safety-critical because misreading signals can lead to severe accidents, such as moving through an intersection during a red light or unexpectedly stopping at a green light. Modern detection systems typically operate in two stages:

- 1) **Traffic-light detection**, which localizes the traffic light in the scene, and
- 2) **Traffic-light classification**, which determines its state (Red, Yellow, Green, or Off).

Although these systems perform well under normal conditions, recent work has shown that deep neural networks are highly vulnerable to adversarial perturbations, which means small modifications that are often imperceptible to humans but significantly alter model outputs [3]. Such perturbations can be digital (pixel-level changes within an image) or physical (stickers or patches placed in the real world). Both forms can lead to dangerous misclassifications in AV perception, particularly in real-time traffic scenarios. Adversarial examples can be generated digitally or physically and present serious safety risks in real-world deployments [2]. While detection tends to be more stable due to the distinguishable shape and structure of traffic lights, classification is far more vulnerable

to visual noise, occlusion, and adversarial manipulation. For this reason, the primary focus of this project is on attacking and improving the robustness of the traffic-light classifier. The detector still identifies the correct bounding box, but adversarial perturbations deliberately attempt to mislead the classifier into predicting the wrong light state.

Previous research has examined such attacks on traffic signs and pedestrian detectors, but comparatively fewer studies focus on the specialized task of traffic light recognition [6].

To address these limitations, this project investigates the vulnerability of the YOLOv8 traffic-light classifier to Expectation-over-Transformation (EOT) adversarial patch attacks, which simulate realistic physical-world variations in lighting, rotation, distance, and viewpoint [2]. We further evaluate whether a simple but effective defense—adversarial patch-based training—can reshape the classifier’s decision boundary and restore robustness under these attacks without compromising real-time performance.

To guide this investigation, the project addresses the following research questions:

- 1) How vulnerable is a YOLOv8-based traffic-light classifier to EOT-optimized adversarial patches designed to induce targeted misclassification (e.g., red $\rightarrow$ green)?
- 2) How does adversarial training with fixed adversarial patches affect the classifier’s robustness, in terms of

classification accuracy, targeted misclassification rate, and detection stability?

- 3) To what extent do EOT patches remain effective across natural scene variations, and how does this compare to the robustness improvements achieved through defense training?

II. BACKGROUND & RELATED WORKS

A. Adversarial Attacks in Deep Learning

Adversarial attacks exploit the linearity and high-dimensional nature of deep neural networks to find perturbations that alter model outputs that are not noticeable to human eyes [3]. In classification settings, a common goal is to craft a perturbation δ such that for a clean image x , the perturbed image $x' = x + \delta$ causes the model to output an incorrect class.

Digital attacks such as the Fast Gradient Sign Method (FGSM) [3] and Projected Gradient Descent (PGD) [5] generate perturbations using gradient information. PGD is regarded as one of the strongest first-order white-box attacks due to its iterative refinement. The PGD update rule is given by:

$$x_{t+1} = \text{Proj}_{\varepsilon} (x_t + \alpha \cdot \text{sign}(\nabla_x L(f(x_t), y))), \quad (1)$$

where x_t is the adversarial sample at iteration t , α is the step size, ε bounds the perturbation magnitude, and L is the loss function. The projection operator $\text{Proj}_{\varepsilon}$ ensures that the perturbation remains within an ε -ball around the clean image. PGD repeatedly updates the input in the direction that increases misclassification while respecting the perturbation constraint.

B. Physical Adversarial Attacks

Physical adversarial examples must remain effective across real-world imaging variations, including changes in view-

point, lighting, and distance. Eykholt et al. [2] introduced the Expectation-over-Transformation (EOT) framework to generate perturbations that survive such variability. Instead of optimizing the loss on a fixed image, EOT optimizes the expected loss over a distribution of transformations T :

$$\min_{\delta} \mathbb{E}_{t \sim T} [L(f(t(x \oplus \delta)), y^*)], \quad (2)$$

where $x \oplus \delta$ denotes overlaying the adversarial patch onto the image, and y^* is the target class for the attack. By optimizing over samples of $t \sim T$, EOT ensures that the adversarial perturbation remains robust under natural distortions encountered in physical environments.

C. Adversarial Robustness

Adversarial robustness refers to the ability of a model to maintain correct predictions in the presence of adversarial manipulations [8]. Models trained exclusively on clean data tend to learn brittle, nearly linear decision boundaries that are easily crossed by small perturbations. Adversarial training improves robustness by introducing adversarial samples during training, thereby reshaping the decision boundary into a more stable, non-linear form that better resists attacks.

D. Traffic-Light Recognition

Most of the prior adversarial research has focused on traffic signs or general object detection. Only a few studies have examined patch-based adversarial attacks targeting the traffic light domain. Munoz et al. [6] showed that misclassification of red lights as green could occur under patch-based adversarial conditions. Sitawarin et al. [9] further explored physical adversarial examples by designing signs that closely resemble real traffic signs but are specifically crafted to mislead autonomous vehicle recognition systems. It shows that these models can be manipulated in physical environments as well.

Later, Pavlitska et al. [7] extended this work by demonstrating realistic, physically deployable adversarial patches that can consistently fool state-of-the-art traffic light detectors under varying lighting and environmental conditions. Zhao et al. [4] introduced AVARA to measure perceptual realism and transferability of physical attacks in simulated environments.

III. EXPERIMENTAL SETUP

A. Dataset

Bosch Small Traffic Lights Dataset (BSTLD) [1] is used for all experiments in this project. This publicly available dataset contains over 13,400 high-resolution images (1280720 pixels) captured from vehicle-mounted cameras on German city streets. It provides more than 24,000 traffic light annotations with four classes: red, green, yellow, off. It is split into training (74%), validation (10%), and test (16%) sets. All annotations were converted into YOLO-compatible format, and images were standardized for training.

B. Baseline Model

The YOLOv8s, a real-time object detector model, is used as a baseline model for the experiments of this project. It is initialized with COCO-pretrained weights and performs bounding-box detection and simultaneously predicts traffic-light state through its classification head. Models were trained on BSTLD with standard augmentations, including brightness shifts, blur, and copy-paste augmentation to mitigate class imbalance.

C. Threat Model

Two threat models are considered for the project that represent both digital and physical adversarial scenarios commonly encountered in autonomous driving.

1) *White-box Digital Adversary (PGD)*: In white box attacks the attacker will have full access to the model's architecture and gradients. PGD is used to optimize the adversarial patch by minimizing the targeted classification loss. The objective is to force red lights to be classified as green.

2) *Black-box Physical Attack (EOT Patch)*: In black-box attack the adversary will not have access to model internals but will be able to modify the physical environment, such as placing printed adversarial patches on or near traffic lights. Patches are trained using EOT to remain effective under variations in rotation, scaling, padding, brightness and translation to simulate real-world deployment.

D. Adversarial Patch Training

After preparing dataset and basemodel setup the adversarial patch training was done. The goal of the adversarial patch training stage is to learn a *universal* patch δ that consistently induces the targeted misclassification of traffic-light states, specifically forcing red lights to be predicted as green. Rather than perturbing the entire image, the patch is optimized so that overlaying it onto the detected traffic-light region alters the classifier's output. The following steps were performed during the training:

1) *Traffic-light Detection*: During patch training, the detector identifies the traffic-light bounding box, and the patch is resized and positioned to cover the lamp head. The optimization process modifies δ so that the classifier assigns the target label y^* (green) whenever the patch is present. A total-variation regularizer is applied to encourage smoothness and physical plausibility. Randomized transformations sampled from T are applied to the patch throughout training to improve its robustness under physical-world view conditions.

2) *PGD-Based Patch Updates*: Patch optimization is further continued using PGD-style iterative updates applied di-



(a) Clean traffic-light image.



(b) Red-light image with adversarial patch applied.

Fig. 1: Visualization of adversarial patch placement. Left: original BSTLD image containing a red traffic light. Right: same image with the optimized EOT-based adversarial patch placed on the traffic-light region.

rectly to the patch tensor. While PGD traditionally perturbs the entire image, here it is used to refine only the patch. Each update step moves the patch in the direction that strengthens its ability to activate YOLOv8’s “green” output. The patch values are projected back into the valid pixel range after each update.

3) *Label-Flipping for Targeted Misclassification*: To enforce the targeted attack, ground-truth labels for red traffic lights are replaced with the target class during training. This label-flipping strategy ensures that the model is explicitly trained to interpret “red + patch” as “green,” to reinforce the desired misclassification behavior.

4) *Bounding-Box-Aware Patch Placement*: Since traffic lights vary significantly in size across the dataset, the patch is scaled relative to the bounding-box dimensions. If a traffic light has width w and height h , the patch size is chosen as a fixed multiple of $\max(w, h)$, ensuring consistent coverage at different viewing distances. The patch is then placed at the center of the detected traffic-light region. This process is illustrated in Fig. 1 by using an example from the BSTLD dataset. The left image shows a clean frame containing a red traffic light. The right image demonstrates how the optimized adversarial patch is resized according to the detected bounding box and placed directly over the lamp head. This visualization

highlights how the patch consistently aligns with the traffic-light region regardless of object size or camera distance, enabling the attack to integrate seamlessly into the scene while remaining physically plausible.

5) *EOT for Physical Robustness*: To approximate real-world deployment, the training incorporates Expectation-over-Transformation (EOT). Each iteration applies a randomly sampled transformation— such as rotation, brightness variation, translation, or slight perspective distortion— before passing the patched image to the model. This process encourages the resulting patch to remain adversarial under the realistic variations encountered by autonomous vehicles.

Finally, PGD-driven updates, label flipping, bounding-box-aware placement, and EOT sampling allows the system to learn a physically robust adversarial patch that reliably induces targeted red-to-green misclassification across diverse scenes.

E. Defense: Adversarial Training with Frozen Patch

To improve robustness against the EOT-based adversarial patch, an adversarial training procedure was applied. Here the optimized patch δ^* from the attack stage is kept frozen and reused during training. For each sample, the detector identifies the traffic-light bounding box, and the frozen patch is scaled and placed on the same region as in the attack stage. The

TABLE I: Evaluation Metrics Under Clean, EOT Patch Attack, and Defense Conditions

Metrics	Clean Data	EOT Patch Attack	Defense (Robust)
Classification Accuracy (%)	87.91	40.77	60.00
Targeted Misclassification Rate (%)	3.12	35.82	20.00
Vanishing TL Rate (%)	9.45	22.31	19.01

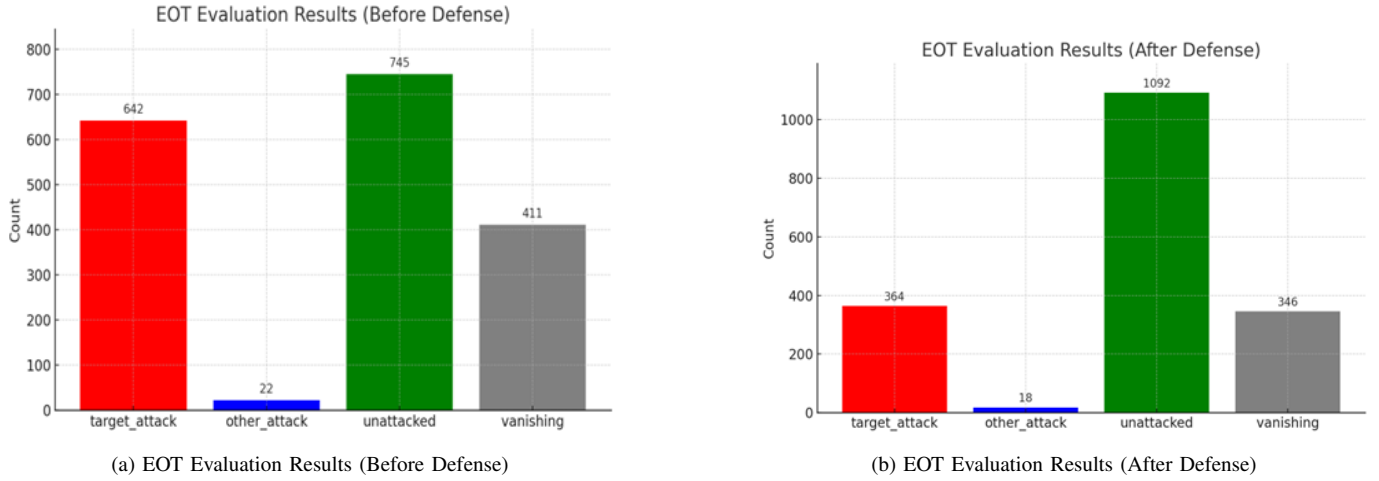


Fig. 2: Comparison of adversarial patch effectiveness before and after adversarial training.

patched image is then passed through the model while keeping the ground-truth labels unchanged, teaching the classifier that the presence of the patch should not alter the traffic-light state. To approximate real-world conditions, random transformations such as rotation, brightness shifts, and small translations are applied through EOT sampling. By repeatedly encountering the patched traffic lights under these variations, the model learns to ignore the adversarial pattern and develops a more stable decision boundary that resists targeted red-to-green misclassification.

IV. RESULTS

This project evaluates the robustness of the traffic-light classifier using three key metrics. Classification Accuracy, Targeted Misclassification Rate (TMR), and Vanishing Rate. The results are shown in Table I. *Classification Accuracy* measures how often the model predicts the correct traffic-light state once the light is detected. *TMR* quantifies how often the attack succeeds in flipping a red light into green, which is the most safety-critical failure. *Vanishing Rate* captures cases

where the traffic light is not detected at all, which means the bounding box disappears under attack. As shown in Table I, for clean data, the model performs reliably by achieving 87.91% accuracy with a very low TMR of 3.12% and a vanishing rate of 9.45%. Under the EOT adversarial patch attack, accuracy drops sharply to 40.77%, and the TMR increases to 35.82%. It shows that the patch is highly effective at forcing red→green misclassification. This is also visible in the Before Defense bar plot in Figure 2a, where the targeted attack count is high (642). After adversarial training with the frozen patch, the model recovers much of its stability by improving accuracy to 60.00%, and decreasing TMR to 20.00%. The vanishing rate also falls slightly to 19.01%. The After Defense Figure 2b reflects this improvement, showing a drop in targeted attacks (364) and a substantial increase in the number of unattacked correct classifications (1092). Overall, the results indicate that while EOT-based patches significantly destabilize the classifier and adversarial training provides meaningful robustness gains.

V. DISCUSSIONS AND LIMITATIONS

The results reveal several important insights about the performance of the traffic-light classifier and highlight challenges that influenced the overall effectiveness of the system. First, the clean-data classification accuracy reached 87.91%, slightly below the expected 90–95% range reported in studies using BSTLD. A major contributing factor is the inherent difficulty of the dataset: traffic lights in BSTLD are very small, often heavily occluded, and exhibit large illumination variability. The use of a lightweight YOLOv8s model, which was chosen for computational feasibility, likely further constrained performance. Larger variants (e.g., YOLOv8m or YOLOv8l) or models incorporating attention mechanisms may have boosted accuracy, but these options were not explored due to GPU memory limitations and training time constraints. Additionally, the training pipeline could not incorporate extensive hyperparameter tuning or advanced augmentations such as glare simulation or multi-scale cropping, which may have improved baseline performance.

The defense results also reveal partial improvement rather than full robustness. Adversarial training reduced targeted misclassification from 35.82% to 20.00%, but did not fully neutralize the adversarial patch. This aligns with known limitations of single-stage adversarial training, which often provides incremental—not complete—robustness. More advanced defenses, such as multi-step patch-aware training or ensemble adversarial defense, could improve resistance but were beyond the project’s computational scope.

Beyond model performance, several broader limitations shaped the final implementation. Although the proposal included experiments on two datasets (BSTLD and LISA), the final system was trained solely on BSTLD. That means cross-dataset generalization and transferability could not be

evaluated. The proposal also described a “patch-aware cropped training” defense, where traffic-light crops would be augmented with synthetic patch variants, but this was not implemented because patch generation, EOT sampling, and full-image training already required substantial computation, which could make additional cropped training pipelines infeasible within the available resources. Similarly, digital PGD attacks were used internally to optimize the adversarial patch but were not evaluated separately as a standalone attack. So a part of the digital-threat space remains unexplored.

VI. CONCLUSION

This project demonstrates that YOLOv8-based traffic-light classification is highly vulnerable to physically realizable adversarial patches generated using the Expectation-over-Transformation (EOT) framework. The optimized patches significantly increase targeted red→green misclassification, posing a serious safety risk for autonomous vehicles. Adversarial training with a frozen patch improves robustness, reducing targeted misclassification from 35.82% to 20.00%, but does not restore clean accuracy nor completely neutralize the attack.

The findings highlight the need for more comprehensive and computationally intensive defense strategies, such as multi-stage adversarial training, temporal reasoning, cross-dataset generalization studies, and hybrid physical–digital robustness techniques. Overall, this work provides a foundation for improving the resilience of traffic-light classification models and emphasizes the importance of addressing adversarial vulnerabilities in safety-critical autonomous driving systems.

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